Customer metrics and firm performance

Loyalty, Churn and Share of Wallet

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Does customer loyalty pay? Two answers

Loyalty pays

Service-profit chain: satisfied employees and customers drive profit

Retention economics: 5% fewer defections, 25% or more profit

Customer equity: retention is the strongest lever on firm value

Loyalty is not enough

Reinartz & Kumar: loyal customers are not always profitable

Keiningham: satisfaction barely moves share of wallet

Ehrenberg-Bass: growth comes from more buyers, not more loyalty

Today: follow the chain from customers to firm value, and test each link against the evidence.

Learning objectives

Define loyalty, churn and share of wallet, and measure each correctly

Explain the service-profit chain and the satisfaction-profit chain

Connect retention and customer equity to firm value and stock returns

Apply CLV, churn targeting and the Wallet Allocation Rule to data

Evaluate where the chain breaks: NPS, loyalty programmes, Double Jeopardy

Concepts and metrics



Part 1 of 6 · what we measure

Three customer outcomes, three questions

Loyalty

Do customers stay and prefer us?

Behavioural: repeat buying

Attitudinal: relative preference

True loyalty needs both (Dick & Basu 1994)

Churn

Do customers leave?

Contractual: cancellation is observed

Non-contractual: silence, so churn must be inferred

Retention rate r = 1 − churn

Share of wallet

How much of their spending do we get?

SOW = our sales ÷ customer's category spend

Needs competitor spend, usually from surveys

Partial defection shows here first

Link to firm value: retention sets how long customers pay, SOW sets how much; together they drive CLV.

Measuring the outcomes

Metric	Formula	Watch out for
Retention rate	customers kept ÷ customers at start	Rises over time as weak customers leave
Expected lifetime	1 ÷ churn (only if all equal)	Underestimates lifetime with heterogeneity
CLV	$m \cdot r \div (1 + d - r)$	Assumes constant margin and retention
Share of wallet	our spend ÷ category spend	Category spend needs survey or panel data
Net Promoter Score	% promoters − % detractors	Not a better growth predictor

Fader & Hardie (2007, 2010); Hardie notes 024, 033, 049 (brucehardie.com); Farris et al. (2021) Marketing Metrics, ch. 2 and 5; Keiningham et al. (2007) Journal of Marketing.

The service-profit chain

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Part 2 of 6 · from employees to profit

The service-profit chain

1	2	3	4	5	6
Internal quality Tools, training, workplace design	**Employees** Satisfaction, retention, productivity	**Service value** Results for customers relative to cost	**Satisfaction** Customers' overall judgement	**Loyalty** Retention, repeat buying, referrals	**Profit** Revenue growth and profitability

The claim: each link drives the next, so investing in employees and service pays off in profit.

Heskett, Jones, Loveman, Sasser & Schlesinger (1994) Harvard Business Review 72(2); Heskett, Sasser & Schlesinger (1997) The Service Profit Chain.

Evidence: the chain holds, with conditions

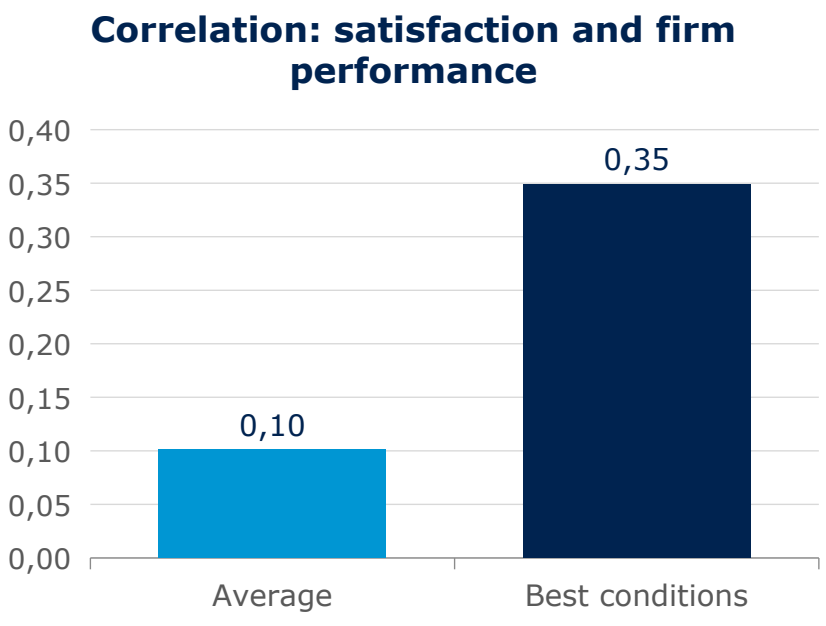
All links — significant in a meta-analysis of the full chain, but effect sizes vary strongly by type of service.
Hogreve, Iseke, Derfuss & Eller (2017) Journal of Marketing 81(3)

+0.5 pts — revenue growth at Sears from +5 units employee attitude, via +1.3 units customer impression.
Rucci, Kirn & Quinn (1998) Harvard Business Review

Time lags — between investment, satisfaction and performance can push managers into myopic cuts.
Evanschitzky, von Wangenheim & Wünderlich (2012) Journal of Retailing

But: in UK grocery stores employee satisfaction did not drive profit (Silvestro 2002), so the employee link is contingent.

Satisfaction pays, but modestly



The satisfaction-profit chain

Asymmetric: a bad experience hurts more than a good one helps

Nonlinear: thresholds and diminishing returns

Intentions ≠ behaviour: same rating, different repurchase rates

Cost side: very high satisfaction can cost more than it earns

Otto, Szymanski & Varadarajan (2020) JAMS, 251 correlations; Anderson & Mittal (2000) JSR; Mittal & Kamakura (2001) JMR; Guenther & Guenther (2021) IJRM; Mittal et al. (2023) Marketing Letters.

From customers to firm value



Part 3 of 6 · alternative frameworks

Six frameworks beyond the chain

Return on quality

Quality is an investment that must pay back; it is possible to overspend.

Rust, Zahorik & Keiningham (1995)

Customer equity

Firm value = sum of customer lifetime values; drivers are value, brand and retention.

Rust, Lemon & Zeithaml (2004); Gupta et al. (2004)

Market assets

Customer relationships speed up, raise and stabilise cash flows.

Srivastava, Shervani & Fahey (1998)

Productivity chain

Actions → customer impact → market impact → financial impact → firm value.

Rust et al. (2004)

Relationships

Trust and commitment keep relationships; relationship quality drives performance.

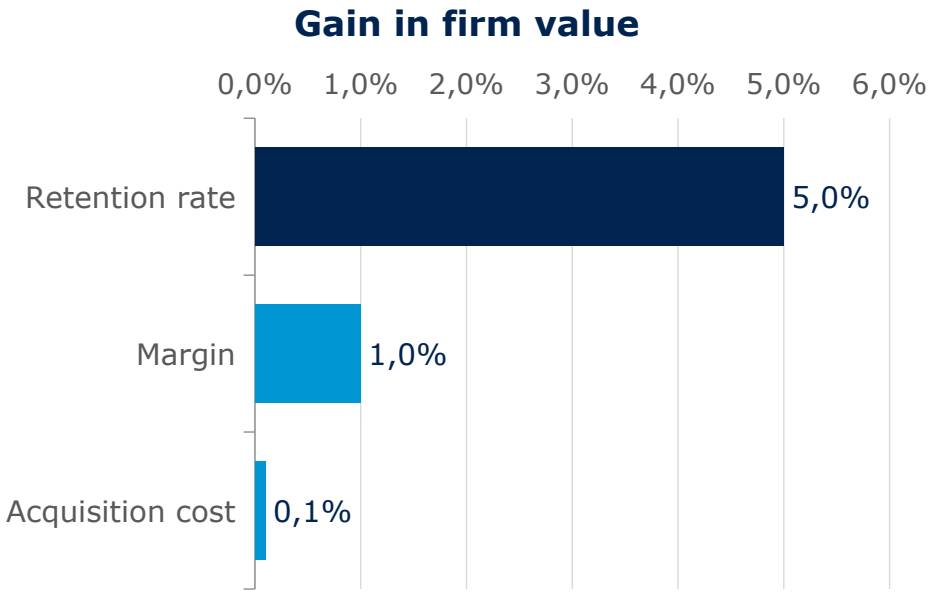
Morgan & Hunt (1994); Palmatier et al. (2006)

Engagement value

CLV plus referral, influence and knowledge value of a customer.

Kumar et al. (2010)

Retention: the strongest value lever



Read with care

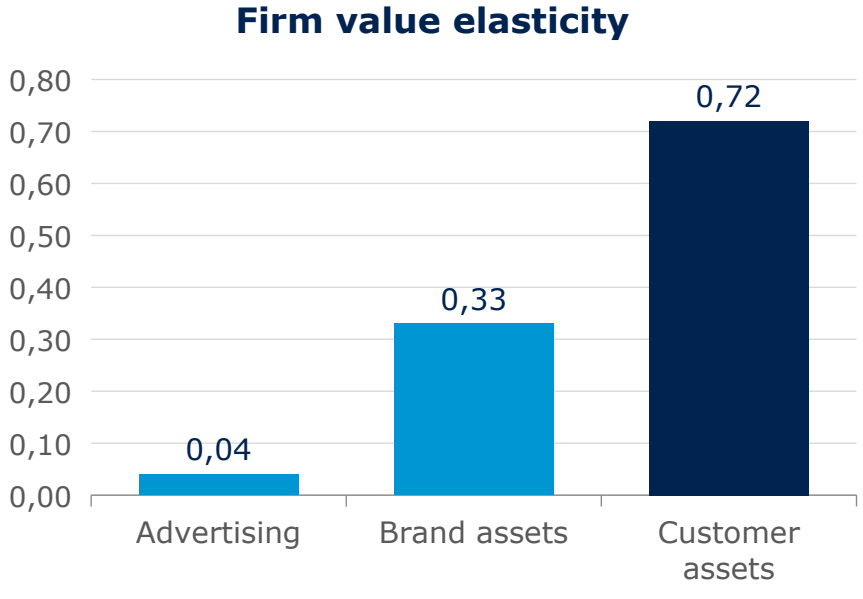
Heterogeneity: a single retention rate understates the effect (Fader & Hardie 2010)

Acquisition: future new customers matter too; leverage 1.55 (Schulze et al. 2012)

Market cap: customer equity predicts it (Kumar & Shah 2009)

Effect of a 1% improvement in each lever. Gupta, Lehmann & Stuart (2004) Journal of Marketing Research 41(1): five firms valued from public data.

Customer assets and the stock market



Satisfaction and returns

Cash flows: more growth, less volatility (Gruca & Rego 2005)

Risk: lower systematic risk (Tuli & Bharadwaj 2009)

Returns: high-ACSI portfolio 518% vs S&P 31%, 2000-2014 (Fornell et al. 2016)

Contested: mispricing limited to few firms (Jacobson & Mizik 2009)

Edeling & Fischer (2016) Journal of Marketing Research: 488 elasticities from 83 studies.

Churn



Part 4 of 6 · predict, then manage

Small churn changes, large value changes

Retention	Churn	CLV	Change
70%	30%	175	base
80%	20%	267	+52%
90%	10%	**450**	**+157%**

Why so steep?

Each point of churn removes a share of every future year's margin

Lifetime rises sharply as churn falls

Formula: CLV = m · r ÷ (1 + d − r), with margin m = 100 and discount rate d = 10%.

Gupta & Lehmann (2005) Managing Customers as Investments; Hardie note 024 on CLV formulas.

Predicting churn is not managing churn

Predict: who will leave?

Methods matter: the churn tournament found gaps worth hundreds of thousands of dollars per campaign

Trees and boosting beat plain logit

Judge by profit, not only AUC

Manage: whom to contact?

Retention futility: the riskiest customers are not the most responsive

Target by uplift, the effect of the offer on each customer

Offers can backfire: churn rose from 6% to 10% after plan recommendations

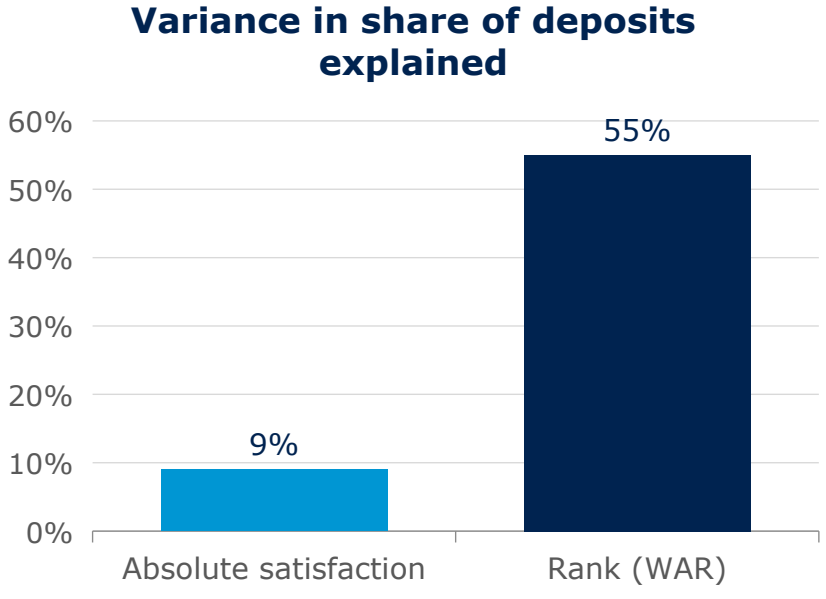
Rule: target customers whose behaviour the offer changes, weighted by their CLV.

Neslin et al. (2006) JMR; Lemmens & Croux (2006) JMR; Verbeke et al. (2012) EJOR; Ascarza (2018) JMR; Ascarza, Iyengar & Schleicher (2016) JMR; Lemmens & Gupta (2020) Marketing Science.

Share of wallet

Part 5 of 6 · rank beats rating

Share of wallet follows rank



≈ 0.1

correlation of changes in satisfaction or NPS with changes in share of wallet

Keiningham et al. (2011) HBR

10×

spending shifts can be worth up to ten times more than outright defections

Coyles & Gokey (2005)

Aksoy (2014) International Journal of Bank Marketing, 4,712 bank customers; Cooil et al. (2007) Journal of Marketing; Keiningham et al. (2015) Journal of Service Management.

The Wallet Allocation Rule in class

Wallet Allocation Rule

$$SOW = (1 - rank \div (n + 1)) \times 2 \div n$$

Rank among 3 brands	Predicted share
1st	**50%**
2nd	33%
3rd	17%

1. Each student lists the supermarkets they used last month
2. Rank them from most to least preferred
3. Estimate the actual split of spending
4. Compare predicted and actual shares in a scatter plot

To grow share, improve your rank against the brands customers also use.

Loyalty programmes: smaller effects

7×

smaller programme effect on share of wallet once self-selection of heavy buyers is controlled.

Leenheer, van Heerde, Bijmolt & Smidts (2007) IJRM

6-9

months until the behaviour change after joining fades again.

Meyer-Waarden & Benavent (2009) JAMS

4.4%

annual growth for high loyalty spenders vs 5.5% for low spenders, 55 listed firms.

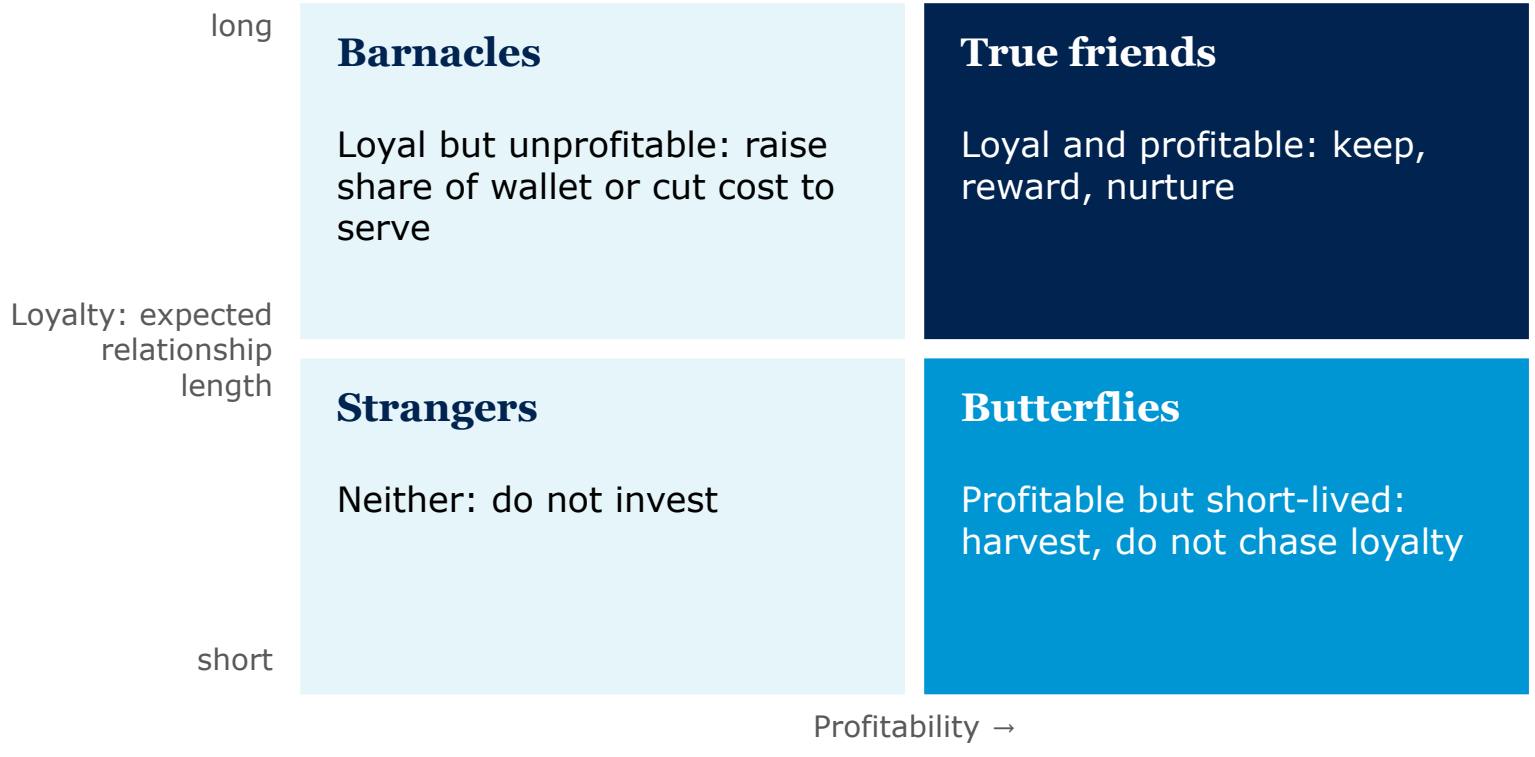
McKinsey (2013) Making loyalty pay

Local cases: jö Bonus Club (4.5 to 4.7m members) and Payback; Starbucks members bring 57 to 60% of US tender.

Critiques and application

Part 6 of 6 · where the chain breaks

Loyalty is not the same as profitability



Reinartz & Kumar (2002) Harvard Business Review; Reinartz & Kumar (2000) Journal of Marketing.

Where the chain breaks

NPS
No better at predicting growth than satisfaction
Keiningham et al. (2007); van Doorn et al. (2013)

Penetration
Brands grow mainly by acquisition; loyalty follows size
Ehrenberg et al. (1990); Riebe et al. (2014)

Cross-buying
Growing wallet share can attract unprofitable buyers
Shah et al. (2012)

Satisfaction cost
Very high satisfaction can lower shareholder returns
Guenther & Guenther (2021)

Employees
Employee satisfaction does not always lift profit
Silvestro (2002)

Lock-in
Switching barriers retain only while experience is poor
Gao, Melero-Polo & Sese (2022)

Lab: churn and CLV in real data

Data | Describe | Model | Value | Decide

Data and tools

IBM Telco: 7,043 customers, churn flag

CDNOW: 2,357 customer sample, purchases

R: CLVTools, BTYDplus (WU), survival

Python: PyMC-Marketing, lifelines

1. Plot retention curves by contract type (Kaplan-Meier)
2. Fit a logit churn model; build a decile lift table
3. Fit BG/NBD on CDNOW; compute P(alive) and annual CLV
4. Value the base: who is worth a retention offer?

Deliverable: one retention chart, one CLV table, half a page on whom to target and why.

Data: Kaggle (IBM Telco Customer Churn); brucehardie.com/datasets (CDNOW). Method: Fader, Hardie & Lee (2005) Marketing Science.

Teaching cases

Pilgrim Bank

Cases (A) and (B), customer-level data: are online customers more profitable and better retained?

Profit, retention, selection bias

HBS 602104 / 602095

Teleko

Choose whom to target with a retention offer: churn risk or uplift?

Churn management

HBS 523005, free online tool

Harrah's

Total Rewards raised share of a gaming budget that started at 36%.

Share of wallet, loyalty data

HBS 502011; Loveman (2003) HBR

Also: Sears (service-profit chain), Starbucks: Delivering Customer Service, Netflix (Darden, CLV and valuation).

References

Core reading

Heskett et al. (1994) Putting the service-profit chain to work, HBR

Gupta & Zeithaml (2006) Customer metrics, Marketing Science

Hogreve et al. (2017) SPC meta-analysis, JM

Gupta et al. (2004) Valuing customers, JMR

Keiningham et al. (2015) Perceptions are relative, JOSM

Frameworks

Srivastava et al. (1998) Market-based assets, JM

Rust et al. (2004) Return on marketing, JM

Reinartz & Kumar (2002) Mismanagement of loyalty, HBR

Otto et al. (2020) Satisfaction meta-analysis, JAMS

Edeling & Fischer (2016) Firm value meta-analysis, JMR

Churn, methods

Ascarza (2018) Retention futility, JMR

Neslin et al. (2006) Defection detection, JMR

Fader et al. (2005) BG/NBD, Marketing Science

McCarthy et al. (2017) Valuing subscription firms, JM

Platzer & Reutterer (2016) Pareto/GGG, Marketing Science

Key takeaways

1. **Metrics:** measure loyalty, churn and share of wallet separately.
2. **Service-profit chain:** every link holds on average, but effects depend on the setting.
3. **Satisfaction:** relates to performance, modestly and nonlinearly.
4. **Retention:** is the strongest lever in customer equity and shows up in firm value.
5. **Share of wallet:** follows a brand's rank against competitors, not its satisfaction score.
6. **Limits:** loyalty is not profit; target churn offers by uplift and CLV.

Next: the churn and CLV lab. Bring a laptop with R or Python installed.

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